

1. PRODUCT AND COMPANY IDENTIFICATION

Product name	: Paraffin (IK/DPK/Laurel)
CAS Identification	: Kerosene
Other means of identification	: Paraffin
Product use	: General purpose fuel
UN number	: 1223
Supplier	: Engen Petroleum Limited (Tel: +27 (0) 21 403 4911, a/h: +27 (0) 21 403 4099)
Poisons Information Helpline	: 0861 555 777 (South Africa)
Spill Response	: 086 100 0366 (South Africa)
Customer Service Centre	: 0860 036 436 (Sales and Technical Information)
Engen Website	: http://www.engen.co.za/

2. HAZARDS IDENTIFICATION

Emergency response data : Colourless Liquid. Flammable. Product can accumulate a static charge and release vapours which may cause a fire or explosion. DOT ERG No. : 128

GHS Classification:

Health

Skin corrosion/irritation	: Hazard Category 2.
STOT - single exposure	: Hazard category 3.
Aspiration hazard	: Hazard Category 1

Environmental

Acute toxicity	: Not classified.
Chronic toxicity	: Hazard category 2.

Physical

Flammability	: Hazard category 3.
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Signal Word : DANGER

GHS Labels/Pictograms:



Hazard Statements

Flammable liquid and vapour. May be fatal if swallowed and enters airways. Causes skin irritation. May cause drowsiness or dizziness. Toxic to aquatic life with long lasting effects.

Precautionary Statements

General

Keep locked up and out of reach of children.

Paraffin (IK/DPK/Laurel)

Prevention

Do not handle until all safety precautions have been read and understood. Use explosion-proof electrical, ventilating and transfer equipment. Ground/bond container and receiving equipment. Use explosion-proof electrical, ventilating and transfer equipment. Use only non-sparking tools. Take precautionary measures against static discharge.

Response

IN CASE OF FIRE: Use dry sand, dry chemical or alcohol-resistant foam to extinguish. IF SWALLOWED: Immediately call a POISON CENTRE or doctor. Do NOT induce vomiting. IF INHALED: Call a POISON CENTRE or doctor immediately. IF ON SKIN (or hair): Immediately remove all contaminated clothing. Gently wash skin with plenty of soap and water. Launder contaminated clothing before re-use.

Storage

Store in a well-ventilated place and keep the container cool and tightly closed.

Disposal

Do not discharge into lakes, streams, ponds and ground water supply.

See Section 11 for further health effects/toxicological data.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name	CAS-No.	Weight%
Kerosene	8008-20-6	100.00

See Section 8 for Exposure Limits (if applicable).

4. FIRST AID MEASURES

Inhalation	:	Move to well-ventilated area. Monitor for respiratory distress; administer oxygen and assist ventilation as required. Seek medical assistance immediately.
Skin contact	:	Remove all contaminated clothing. Wash area with soap and water for 10 to 15 minutes.
Eye contact	:	Remove contact lenses and irrigate exposed eyes for at least 15 minutes. Irrigate before and after removing the lenses to prevent a carry-over of the substances to the shielded area of the lens.
Ingestion	:	Immediately call a POISON CENTER or doctor/ physician. Do NOT induce vomiting. If vomiting does occur, have victim lean forward to reduce risk of aspiration.

5. FIRE-FIGHTING MEASURES

Extinguishing media	:	Carbon dioxide, foam, dry chemical and water fog.
Special fire fighting procedure	:	Water spray should only be used to keep fire-exposed containers cool, flush spills away from exposures, disperse vapours and protect personnel attempting to stop leak. Prevent runoff from fire control or dilution from entering streams, municipal sewers, or drinking water supply.
Special protective equipment for firefighters	:	For fires in enclosed areas, fire fighters must use Self-Contained Breathing Apparatus.

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Unusual fire and explosive hazards	:	Flammable.
Products of decomposition	:	Fumes, smoke and carbon monoxide.
Flash Point	:	$\geq 38\text{ }^{\circ}\text{C}$ (ASTM D-56)
Upper Explosion Limit (UEL)	:	5.0 %(V)
Lower Explosion Limit (LEL)	:	0.7 %(V)
NFPA Hazard Id	:	Health: 0; Flammability: 2; Reactivity: 0

6. ACCIDENTAL RELEASE MEASURES

Procedure if material is released or spilled	:	Report spills/releases as required to appropriate authorities.
Methods for cleaning up	:	Absorb on fire retardant treated sawdust, diatomaceous earth, etc. Shovel up for later approved disposal. LARGE SPILLS: Contain material and pump back to holding tank for disposal or recycling in accordance with product characteristics and applicable regulations.
Personal precautions	:	See Section 8.
Environmental precautions	:	Prevent spills from entering municipal sewers or drains and contact with soil.

7. HANDLING AND STORAGE

Safe handling advice	:	Harmful in contact with or if absorbed through the skin. Avoid inhalation of vapours or mists. Use in well ventilated area away from all ignition sources.
Storage information	:	Store in a cool area. Small containers of approved design, properly sealed and labeled, should be stored in well ventilated surroundings and kept out of reach of children.
Storage and handling procedures	:	To minimize the risk of fire or explosion from discharges, static and/or vapour accumulation, effectively bond and ground product storage and transfer systems. Electrical equipment and fittings must comply with local fire prevention regulations for this class of product. Refer to national or local regulations covering safety at petroleum handling and storage areas for this product.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OELs)

Components	CAS-No.	Source	TWA	Value		Notations
Kerosene	8008-20-6	ACGIH	LTEL	200 mg/m ³	50 ppm	Skin; A3; BEI

LTEL: Long Term Exposure Limits - Time Weight Average (TWA) over 8 hours.

STEL: Short Term Exposure Limits - Time Weight Average (TWA) over 15 Minutes

Note: Limits Shown for guidance only. Follow applicable regulations.

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Personal Protective Equipment (PPE)

Engineering controls	:	Use in well ventilated area. Explosive-proof ventilation equipment with local exhaust is desirable.
Respiratory protection	:	Approved respiratory equipment must be used when airborne concentrations are unknown or exceed the recommended exposure limit. Self-Contained Breathing Apparatus may be required for use in confined or enclosed spaces.
Eye protection	:	If splash with liquid is possible, chemical type goggles should be worn.
Skin and body protection	:	Impervious gloves must be worn. If body contact is likely, appropriate personal protective equipment must be worn. Good personal hygiene practices should always be followed.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	:	Liquid.
Colour	:	Colourless
Odour	:	Mild
Solubility	:	Negligible
Boiling point	:	> 140 °C
Flash Point	:	>= 38 °C (ASTM D-56)
Upper Explosion Limit (UEL)	:	5.0 %(V)
Lower Explosion Limit (LEL)	:	0.7 %(V)
Vapour pressure	:	3 hPa
Density	:	0.79 g/cm ³ @ (ASTM D-4052)
Viscosity, kinematic	:	1.5 mm ² /s @ 40 °C (ASTM D-445)

10. STABILITY AND REACTIVITY

Stability	:	Stable.
Conditions to avoid	:	Heat, sparks, flame and build up of static electricity.
Materials to avoid	:	Strong oxidizers.
Hazardous decomposition products	:	Fumes, smoke and carbon monoxide.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Components

Kerosene

Acute oral toxicity	:	LD50 (Rats) : > 5,000 mg/kg
Acute inhalation toxicity	:	LC50 (Rats) : > 5.28 mg/l
Acute dermal toxicity	:	LD50 (Rats) : > 2,000 mg/kg

Skin corrosion/irritation

Components

Kerosene

Result	:	Causes skin irritation.
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Eye irritation Components

Kerosene
Result : No known significant effects or critical hazards.

Sensitization Components

Kerosene
Remarks : This product is not a skin sensitizer.

Germ cell mutagenicity Components

Kerosene
Remarks : No known significant effects or critical hazards.

Carcinogenicity Components

Kerosene
Remarks : Not classifiable as a human carcinogen.

Reproductive toxicity Components

Kerosene
Remarks : No known significant effects or critical hazards.

STOT - single exposure Components

Kerosene
Remarks : May cause drowsiness or dizziness.

Specific target organ toxicity (STOT) - repeated exposure Components

Kerosene
Remarks : No known significant effects or critical hazards.

Aspiration hazard Components

Kerosene
Remarks : May be fatal if swallowed and enters airways.

12. ECOLOGICAL INFORMATION

Ecotoxicity effects Components

Kerosene
Acute toxicity : Not established.
Chronic toxicity : Fish, invertebrates and algae with LC/EC50 >1 - 10 mg/L

Persistence and degradability Components

Kerosene

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Biodegradability : This product is expected to be inherently biodegradable.

Bioaccumulation Components

Kerosene
Bioaccumulation : Log Kow > 4 (potentially bioaccumulative)

13. DISPOSAL CONSIDERATIONS

Waste disposal : Product is suitable for burning for fuel value compliance with applicable laws and regulations.

Contaminated packaging : Empty containers retain residue (liquid and/or vapour) and can be dangerous. DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH. Do not attempt to refill or clean container since residue is difficult to remove. Empty drums should be completely drained, properly bunged and promptly returned to a drum reconditioner. All containers should be disposed of in an environmentally safe manner and in accordance with governmental regulations.

Other regulations : Disposal of unused product may be subject to RCRA regulations (40 CFR 261). Disposal of the used product may also be regulated due to ignitability, corrosivity, reactivity, or toxicity as determined by the Toxicity Characteristic Leaching Procedure (TCLP).

Flash Point : $\geq 38\text{ }^{\circ}\text{C}$ (ASTM D-56)

14. TRANSPORT INFORMATION

Note : The flash point of this material is $> 38\text{ }^{\circ}\text{C}$, and hence regulatory classifications for flammability may vary.

ADR

Proper shipping name : KEROSENE
UN number : 1223
Class : 3
Letter : F
Packing group : III
Labelling number : 3

CFR

Proper shipping name : KEROSENE
UN number : 1223
Class : 3
Letter : F
Packing group : III
Labelling number : 3

IATA_C

Proper shipping name : KEROSENE
UN number : 1223
Class : 3
Letter : F
Packing group : III

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Labelling number : 3

IMDG

Proper shipping name : KEROSENE
UN number : 1223
Class : 3
Letter : F
Packing group : III
Labelling number : 3
Marine pollutant : Marine Pollutant
Medical First Aid Guide (MFAG) table : 311
Emergency Schedule (EmS) number : 3-07
IMDG code page number : 3375

15. REGULATORY INFORMATION

South African Legislation and Standards : South African OHS Act, 85 of 1993: Hazardous Chemical Substances, Regulation 9A
National Environmental Management: Waste Act 2008

South African Guidelines
SANS 10234:2008 - Globally Harmonized System of classification & labelling of chemicals
SANS 11014:2010 - Safety data sheet for chemical products - Content & order of sections

Other regulations : Globally Harmonized System of Classification and Labelling of Chemicals (GHS) [GHS (Rev. 7) (2017)]

16. OTHER INFORMATION

INJECTION INJURY WARNING: If product is injected into or under the skin, or into any part of the body, regardless of the appearance of the wound or its size, the individual should be evaluated immediately by a doctor as a surgical emergency. Even though initial symptoms from high pressure injection may be minimal or absent, early surgical treatment within the first few hours may significantly reduce the ultimate extent of injury.

Abbreviations That May Have Been Used In This Document:

TLV - Threshold Limit Value	TWA - Time Weighted Average
LTEL - Long-term Exposure Limit	STEL - Short-term Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists	CAS - Chemical Abstract Service Number
IMO/IMDG - International Maritime Dangerous Goods Code	SDS – Safety Data Sheet
SANS – South African National Standards	GHS - Globally Harmonized System of Classification and Labelling of Chemicals
STOT – Specific Target Organ Toxicity	OHS Act – Occupational Health and Safety Act
NFPA - National Fire Protection Association (USA)	ECHA - European Chemicals Agency

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Sources of key data used to compile the Safety Data Sheet	:	European Oil Company Organisation for Environment, Health and Safety) ECHA (https://echa.europa.eu/information-on-chemicals)
Original Release	:	27.09.2005
Last Date of Review	:	31.12.2019

Disclaimer

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